Given Γ in G with parameterization $\boldsymbol{q}(t)$, $t \in [\alpha, \beta]$, we consider the curve Γ^ε with parameterization $\boldsymbol{q}^\varepsilon$ defined by

$$\boldsymbol{q}^\varepsilon(t) = \boldsymbol{q}(t) + \varepsilon\boldsymbol{\eta}(t), \qquad t \in [\alpha, \beta], \tag{2.3}$$

where $\boldsymbol{\eta}(t)$ satisfies $\boldsymbol{\eta}(\alpha) = \boldsymbol{\eta}(\beta) = 0$. Thus $\boldsymbol{\eta}$ is a C^∞ parameterized curve linking $\mathbf{0}$ to $\mathbf{0}$. In defining variations in this way, we are using, implicitly, the fact that we can add together functions in C^∞ and multiply them by scalars (e.g. ε) and remain in C^∞. We also make use of the intuitive concept that (2.3) defines $\boldsymbol{q}^\varepsilon$ in such a way that it is "close" to $\boldsymbol{q}$ when ε is small. This can be made precise by a little more elaboration, but we forego this here. Effectively, we are using our understanding of C^∞ as a *normed function space* to restrict attention to variations of the base curve Γ in a particular "direction."

Using a Taylor series expansion of the Lagrangian,[3] we have

$$\begin{aligned}\mathcal{A}_L(\Gamma^\varepsilon) - \mathcal{A}_L(\Gamma) &= \int_\alpha^\beta [L(\boldsymbol{q}(t) + \varepsilon\boldsymbol{\eta}(t), \dot{\boldsymbol{q}}(t) + \varepsilon\dot{\boldsymbol{\eta}}(t)) - L(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t))]\,\mathrm{d}t,\\ &= \int_\alpha^\beta \left[\varepsilon\left(\frac{\partial L}{\partial \boldsymbol{q}}(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t))\boldsymbol{\eta}(t) + \frac{\partial L}{\partial \dot{\boldsymbol{q}}}(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t))\dot{\boldsymbol{\eta}}(t)\right) + \mathcal{O}(\varepsilon^2)\right]\mathrm{d}t.\end{aligned}$$

Hamilton's principle states that the natural motion of the system described by the Lagrangian L is a stationary point of the classical action which implies that the $\mathcal{O}(\varepsilon)$ term above should vanish for any smooth variation $\boldsymbol{\eta}(t)$ with $\boldsymbol{\eta}(\alpha) = \boldsymbol{\eta}(\beta) = 0$, i.e.,

[3] In the multidimensional setting, Taylor's theorem states that given a C^{k+1} function $f : \mathbb{R}^m \to \mathbb{R}$ and a point z_0, we have

$$\begin{aligned}f(z) - f(z_0) = \nabla f(z_0)\cdot(z - z_0) + f^{(2)}\langle z - z_0, z - z_0\rangle + f^{(3)}\langle z - z_0, z - z_0, z - z_0\rangle\\ + \ldots f^{(k)}\langle z - z_0, z - z_0, \ldots, z - z_0\rangle + \mathcal{O}(\|z - z_0\|^{k+1})\end{aligned}$$

where $\nabla f = \partial f/\partial z$ is the gradient, i.e. a vector with m components, $f^{(2)}$ is the $m \times m$ Hessian matrix of f (the matrix whose ij component is $\partial^2 f/\partial z_i \partial z_j$), and $f^{(2)}\langle \boldsymbol{u}, \boldsymbol{v}\rangle$, $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^m$, represents the quadratic form $\boldsymbol{u}^T f^{(2)}\boldsymbol{v}$. In a similar way we interpret $f^{(3)}$ as a tensor which we can think of as a $m \times m \times m$ triply-indexed array, the ijk element being $\partial^3 f/\partial z_i \partial z_j \partial z_k$ and

$$f^{(3)}\langle \boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}\rangle = \sum_{i=1}^m \sum_{j=1}^m \sum_{k=1}^m (\partial^3 f/\partial z_i \partial z_j \partial z_k) u_i v_j w_k.$$